



Tapped Hardest, Fixed First

Auditing Whether Refill-Counter Reads Predict a Water-Fountain's Repair Priority

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Motivation

Facilities ranks the University's water-bubblers for repair by digital refill-counter reads, on the assumption that heavy tapping signals both wear and urgency. Whether the counter tracks fountain condition, or simply which taps sit on the busiest walking route, has not been tested.

Objectives

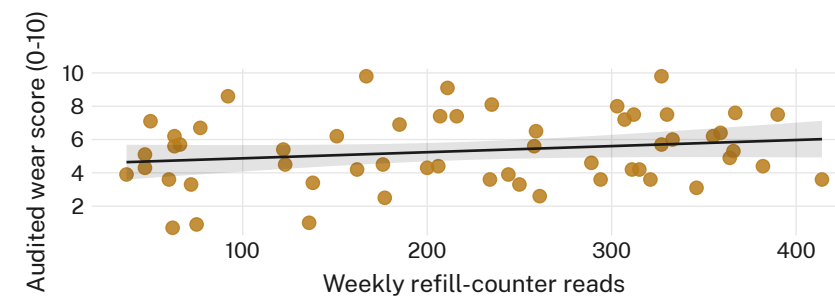
- Compare refill-counter reads against an independent condition audit of the same fountains
- Test whether counter rank predicts repair-queue position
- Identify what the counter reads, if not condition

Approach

- 56 bubblers · fortnightly counter reads · 22-week window (to 3 July 2026)
- Blind condition audit: flow rate, chill delay, basin fouling scored by assistants unaware of counter rank
- Repair-queue position drawn from Facilities' ticketing log at audit close

Findings

Refill-counter reads explained little of the variance in independently audited wear or in eventual queue position; the fountains serviced first sat mostly on the busiest lecture-changeover routes.



Refill-counter reads explained 4% of the variance in independently audited wear ($R^2 = 0.04$, $n = 56$) across the 22 weeks to 3 July 2026; the strongest predictor of repair-queue position was walking distance to the nearest lecture-theatre changeover.

Interpretation & conclusion

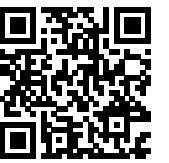
A refill count rewards footfall and timetable rhythm, not condition. Facilities' repair queue currently privileges the busiest hour over the driest basin. Next: trial a wear-weighted queue sampled outside changeover peaks.

Further reading

1. Strathern, M. (1997). 'Improving ratings': audit in the British University system. *European Review*, 5(3), 305–321. doi:10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4
2. Hood, C. (2006). Gaming in Targetworld: The targets approach to managing British public services. *Public Administration Review*, 66(4), 515–521. doi:10.1111/j.1540-6210.2006.00612.x
3. Ahmad, R., & Kamaruddin, S. (2012). An overview of time-based and condition-based maintenance in industrial application. *Computers & Industrial Engineering*, 63(1), 135–149. doi:10.1016/j.cie.2012.02.002
4. Zonta, T., da Costa, C. A., da Rosa Righi, R., de Lima, M. J., da Trindade, E. S., & Li, G. P. (2020). Predictive maintenance in the Industry 4.0: A systematic literature review. *Computers & Industrial Engineering*, 150, 106889. doi:10.1016/j.cie.2020.106889
5. Fire, M., & Guestrin, C. (2018). Over-optimization of academic publishing metrics: Observing Goodhart's Law in action. arXiv:1809.07841.

Condition audit blind to counter rank; thresholds pre-registered; no fountain taken offline beyond its usual maintenance window.

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“The thirstiest tap isn’t always the driest fountain.”

Refill-counter rank explained 4% of audited wear across 56 campus bubblers.